

When does an LLM trust a specialist model? A cost-aware trust-routing audit — and why “calibrated reliability interfaces” are not a free win

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Abstract

Frontier AI-for-science systems increasingly place a general LLM *above* specialist scientific models (protein, genome, single-cell), where the LLM must decide — under a verification cost — whether to **trust** a specialist’s output, **verify** it, fall back to a **cheap baseline**, or **defer**. We audit that trust-routing decision and ask whether converting a specialist’s reliability evidence into an explicit *calibrated reliability interface* improves routing over simply showing the raw confidence.

Across three substrates (a GEARS/Norman perturbation dry-run, a single-cell foundation-model cue, and protein-structure prediction with Boltz-2/pLDDT) the robust finding is **cautionary, not celebratory**. (1) The LLM reasoning layer is strongly **cue-sensitive**: shown any reliability signal it routes far better than with none, but it also follows misleading signals. (2) On a *validated, calibrated* signal (pLDDT, which predicts true lDDT at Pearson 0.89 for monomers), an LLM (Claude Sonnet 4.6) converts it into much better cost-aware routing — **but the benefit comes from the calibrated *information*, not a directive recommendation, and packaging it as a “reliability card” does not robustly beat showing the raw confidence number** ($\Delta_{\text{net}} +0.025$, 95% CI $[-0.037, +0.087]$). (3) The card is **model-dependent and can backfire**: Claude Opus 4.8, a more risk-averse model, **over-verifies** under the card (97.5% verify rate) and is *significantly worse* than with raw confidence ($\Delta_{\text{net}} -0.225$, CI $[-0.325, -0.113]$), replicating the “stronger reasoning $\neq$ better cost-aware orchestration” pattern. We conclude that **raw calibrated confidence is the robust lever; prompt-visible reliability *interfaces* are not a free win and are model-dependent** — presentation alone is insufficient, motivating enforcement-based (tool/MCP/post-training) routing.

1. Question

What does an LLM reasoning layer trust when it reads a specialist scientific model’s output, and can that trust be made calibrated, auditable, and useful under verification cost?

Action set per item: `trust_sfm` | `verify_assay` | `default_baseline` | `defer`. Reward: `net = correct - lambda*assays` (λ = verification price; primary $\lambda = 0.5$).

2. What each phase found

- **Phase 0 (GEARS/Norman dry run, 642 non-leakage Sonnet requests).** Routing is strongly cue-sensitive and only weakly calibrated; additive-disagreement evidence helps, misleading “reliability cards” hurt, and self-reported rationales frequently cite unavailable evidence.
- **Phase 1 (single-cell scFoundation cue).** NO-GO. The scFoundation-derived edge cue predicts GEARS wrongness at AUROC 0.599 vs 0.576 for a random same-gene control — near-noise. A specialist-metric check found GEARS $\approx$ a cheap additive baseline (replicating Ahlmann-Eltze et al. 2025). You cannot study calibrated trust where the signal carries no validated information.
- **Phase 2 (protein structure, Boltz-2 / pLDDT).** A substrate where the specialist is excellent *and* emits a validated calibrated confidence — the focus of this report.

3. Phase 2 methods (brief)

- **Substrate.** 80 RCSB targets released 2026-06-17 (after Boltz-2’s 2023-06 training cutoff $\rightarrow$ leakage-safe), 40 monomer + 40 complex; sequences + reference structures from RCSB; structures predicted with Boltz-2 on an A100/A40 (Cayuga).
- **Truth.** Superposition-free CA-IDD_T of prediction vs experimental reference (Mariani et al. 2013), monomer single-chain and complex all-chain (interface-inclusive). *Caveat: home-grown CA-IDD_T, not yet validated against OpenStructure; complexes use IDD_T, not DockQ.*
- **Calibrated signal.** pLDD_T $\rightarrow$ P(wrong) via leave-one-out isotonic regression. pLDD_T predicts IDD_T at **Pearson 0.89 (monomers), 0.16 (complexes)** — the designed monomer $\rightarrow$ complex calibration gap, confirmed.
- **Offline gate.** A deterministic “verify iff calibrated-risk $> \lambda$ ” policy must beat trust-all + shuffled/inverted controls before any LLM call. *Caveat: it is degenerate at the standard IDD_T ≥ 0.7 cutoff (Boltz-2 is correct on $\sim 95\%$ of recent targets) and was made to pass at IDD_T ≥ 0.9 — a post-hoc choice; signal AUROC there is 0.89.*
- **LLM pilot.** A balanced **40-target subset (20 monomer + 20 complex)** of the 80-target benchmark, stratified across the IDD_T range, $\times 5$ arms = 200 requests, Sonnet 4.6 and Opus 4.8, scored against held-out IDD_T-truth. (The full 80 targets are used for the offline calibration gate.) Arms: `no_signal`, `raw_plddt_shown`, `calibrated_risk_shown_no_recommendation` (information without a directive), `calibrated_interface_s`, `inverted_reliability_interface_control` (risk + recommended action), `inverted_reliability_interface_control`.

4. Results

4.1 Sonnet ($\lambda = 0.5$, IDD_T ≥ 0.9 ; target-bootstrap 95% CIs, seed 13)

arm	net/target	Δ vs no_signal (95% CI)
calibrated_interface_shown	0.738	+0.487 [+0.338, +0.625]
calibrated_risk_shown_no_recommendation	0.725	+0.475 [+0.325, +0.613]
raw_plddt_shown	0.713	+0.463 [+0.300, +0.613]
inverted_reliability_interface_control	0.388	+0.138 [+0.025, +0.250]
no_signal	0.250	—

- **Any confidence cue robustly beats no-signal** (+0.46–0.49, CIs exclude 0). With no

signal Claude defers/over-verifies (“can’t assess reliability”).

- **A4 — the benefit is informational, not directive.** calibrated_interface vs calibrated_risk_no_recommendation: $\Delta +0.013$, CI $[+0.000, +0.037] \approx 0$.
- **The calibrated card does NOT robustly beat raw pLDDT.** calibrated_interface vs raw_plddt: $\Delta +0.025$, CI $[-0.037, +0.087]$ — crosses zero.
- **Regime-appropriate caution.** Under the card, verify rate is 0.55 on complexes vs 0.20 on monomers — more checking exactly where pLDDT calibration degrades.
- **Cue-sensitivity persists.** The inverted card degrades routing $0.738 \rightarrow 0.388$.

4.2 Cross-model (Opus 4.8, same 200 requests)

arm	Sonnet net	Opus net	Sonnet verify	Opus verify
raw_plddt_shown	0.713	0.713	0.225	0.275
calibrated_interface_shown	0.738	0.487	0.375	0.975
no_signal	0.250	0.150	0.500	0.300

Opus calibrated_interface vs raw_plddt: $\Delta -0.225$, CI $[-0.325, -0.113]$ — **robustly worse**. The risk-framed card triggers blanket over-verification in the more risk-averse model (it verifies even confident monomers). A4 holds for Opus too (card vs no-recommendation = 0).

4.3 Robustness

The Sonnet/Opus conclusions are **stable across IDDT cutoffs 0.5–0.9** (results/phase2_preflight/cutoff_results). calibrated_interface — raw is ≈ 0 for Sonnet and ≈ -0.25 for Opus at every cutoff — so the post-hoc 0.9 choice affects only the offline-gate variance, not the pilot conclusions. Across $\lambda = 0.2/0.5/0.8$ the lower arms keep their order; the only change is that calibrated_interface and raw_plddt swap at the top under high cost (card—raw: $+0.070 / +0.025 / -0.020$ at $\lambda = 0.2/0.5/0.8$) — which further supports that the card does not robustly beat raw confidence.

5. Interpretation

The robust, cross-phase, cross-model claim:

The LLM reasoning layer is **cue-sensitive**; surfacing **raw calibrated confidence** is the robust lever for better cost-aware routing; repackaging it as a calibrated **reliability interface** is **not a free win and is model-dependent** — neutral for Sonnet and harmful for the more risk-averse Opus, which over-verifies. The benefit, where present, is **informational, not directive**. Presentation alone is insufficient; **enforcement** (tools, MCP constraints, or post-training) is the natural next lever.

This extends Turpin et al. (2023) — LLMs follow cues regardless of validity — into a constructive, *calibrated-signal* regime: a correctly calibrated cue helps (via its information), a misleading one hurts, and a “reliability card” framing can over-trigger a risk-averse model.

6. Relation to prior work

- **Proto / EvoDesign** (Merchant, Guo, Viggiano, ... Hie; bioRxiv 2026.06.22.733870) — **the most directly relevant system**. Proto is a high-level programming language

for generative biology: an LLM/agent composes ~ 120 specialist models (Evo2, ESM3, AlphaFold3, Boltz-2, AlphaGenome, ProteinMPNN, ...) into multi-objective design programs, treating each specialist’s confidence as an energy to minimize ($\pi(x) \sim p(x)\exp(-f(x)/T)$). *It is the strongest existing validation of the hybrid “reasoning layer above specialist models” stack this project assumes – but it trusts specialist confidence UNCONDITIONALLY: no calibrated trust/verify/defer decision, no verification price, no audit of the reasoning layer’s trust. Proto itself reports the gap we study – “structure-prediction metrics are plausibility filters, not guarantees of function”; “no single in-silico metric separated functional from non-functional designs” – and routes around it with multi-objective consensus plus tens of wet-lab tests. Two of its constructions are concrete, uncalibrated* precedents for our actions:* (i) its 8-stage CRISPR filter cascade runs cheap filters first and expensive AlphaFold3 last with short-circuiting – a hand-built, cost-ordered verify-vs-defer pipeline (our net = correct - lambda*assays objective, minus the calibration); (ii) its multi-oracle structure consensus (require Boltz-2 AND AF2 AND AF3 to agree) is an uncalibrated verify-by-consensus. Our contribution is the calibrated, cost-aware, pre-registered trust layer that Proto-class agentic systems currently lack.

- **Cue-following / unfaithfulness:** Turpin et al. 2023. We add a cost-aware routing game and a *validated-calibrated* cue.
- **Cost-aware routing / selective prediction:** ASPEST (Kim et al. 2023), Adaptive-RAG, FrugalGPT/RouteLLM, learning-to-defer (Madras 2018; Mozannar & Sontag 2020). We route over a *fallible scientific specialist* under an assay cost.
- **Specialist soundness:** Ahlmann-Eltze, Huber & Anders 2025 (perturbation FMs $\approx$ linear baselines — why single-cell failed here); Jumper et al. 2021 and the McGuffin-group calibration audit 2024 (pLDDT calibration — why protein structure worked); Cheng et al. 2023 (AlphaMissense — a runner-up substrate).

7. Limitations

- Offline-gate PASS engineered post-hoc (IDDT ≥ 0.9 + in-distribution LOO calibration);
- home-grown CA-IDDT, unvalidated vs OpenStructure, complexes not DockQ;
- low-stakes substrate ($\sim 95\%$ correct at IDDT ≥ 0.7);
- the model-visible packet contains only confidence (+ regime), so “uses the signal” is partly forced;
- $n = 40$, single primary λ , single MSA source, one run per model, default-False template baseline. A pre-registered confirmatory redo addressing all of these is specified in `PHASE2_PREREGISTRATION.md`.

7b. Post-publication validation (2026-06-23, on-cluster)

Three follow-up checks on Cayuga directly test the limitations above against gold-standard tools (artifacts under `results/phase2_{ost,dockq,leakage}_*`):

- **Monomer truth — confirmed.** OpenStructure all-atom IDDT vs the home-grown CA-IDDT on the 40 monomers: Pearson **0.99**; pLDDT $\rightarrow$ OST-IDDT = **0.894**, confirming the headline 0.89 monomer calibration against the gold standard.
- **Complex truth — corrected.** The all-chain CA-IDDT is intra-chain-dominated, so the reported pLDDT $\rightarrow$ complex “calibration collapse” (0.16) was largely a **metric artifact**. Against gold-standard **DockQ** (40 complexes): pLDDT $\rightarrow$ DockQ = **0.44**, and the proper interface confidence **ipTM $\rightarrow$ DockQ** = **0.77** — the specialist *is* well-calibrated on complexes; route on ipTM, not pLDDT. Re-scoring the v1 episodes with DockQ truth leaves the headline intact (interface-raw ≈ 0 for Sonnet, ≈ -0.25 for Opus).

- **Leakage — the “leakage-safe” claim does not hold.** An MMseqs2 search of all 80 targets vs the full PDB, date-filtered via RCSB, finds that **~70% (56/80 at a 2024 cutoff) have a \$ \$90%-identical pre-cutoff homolog**. Release-date curation did not prevent training leakage; the ~95% base rate is partly memorization. The pre-registered redo’s homolog dedup is therefore empirically required, and the substrate must be re-curated for genuinely low-homology folds.

Net: the routing **headline survives**, the monomer metric holds, the complex-calibration story is corrected, and the substrate’s leakage is the principal reason to execute the confirmatory redo rather than lean on v1.

8. Reproducibility

All code, tests (163 passing), and compact result artifacts are in the repo under `experiments/trust_cue_attribute`. Key artifacts: `results/phase2_preflight/{boltz_contract,boltz_smoke_report,calibration_gate,calibrated_gate,interface_pilot_score,interface_pilot_robustness, interface_pilot_score_opus,interface_pilot_score_opus_opus}`. Pipeline: `phase2_curate_pdb.py` → Boltz predict → `phase2_truth.py` (CA-IDDT) → `phase2_records.py` → `phase2_calibrated_gate.py` → `phase2_interface_pilot.py` → `phase2_score_episodes.py` / `phase2_robustness.py`. Large JSONL (requests, episodes, structures) stay ignored under `hpc_outputs/`.

How to cite

See `CITATION.cff`. This report is intended for archival via Zenodo (GitHub release → DOI); the pre-registration is intended for OSF.